

LAB 5

- 1) Screenshot of detailed information about bind9 package in apt (10 points)

```
cpres230@ns1:~$ apt show bind9
Package: bind9
Version: 1:9.18.30-0ubuntu0.22.04.2
Priority: optional
Section: net
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Debian DNS Team <team+dns@tracker.debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 988 kB
Pre-Depends: init-system-helpers (>= 1.54~)
Depends: adduser, bind9-libs (= 1:9.18.30-0ubuntu0.22.04.2), bind9-utils (= 1:9.18.30-0ubuntu0.22.04.2), debconf | debconf-2.0, dns-root-data, iproute2, lsb-base (>= 3.2-14), netbase, libc6 (>= 2.34), libcap2 (>= 1:2.10), libjson-c5 (>= 0.15), libltdb0 (>= 0.9.7), libmaxminddb0 (>= 1.3.0), libnhttp2-14 (>= 1.3.0), libssl3 (>= 3.0.0~alpha1), libuv1 (>= 1.4.2), libxml2 (>= 2.7.4), zlib1g (>= 1:1.14)
Suggests: bind-doc, dnsutils, resolvconf, ufw
Breaks: bind (<< 1:9.13.6~)
Replaces: bind (<< 1:9.13.6~)
Homepage: https://www.isc.org/downloads/bind/
Task: dns-server
Download-Size: 261 kB
APT-Sources: http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages
Description: Internet Domain Name Server
 The Berkeley Internet Name Domain (BIND 9) implements an Internet domain
 name server. BIND 9 is the most widely-used name server software on the
 Internet, and is supported by the Internet Software Consortium, www.isc.org.
.
 This package provides the server and related configuration files.

N: There is 1 additional record. Please use the '-a' switch to see it
cpres230@ns1:~$ _
```

- 2) Screenshot of debsums command (10 points)

```
cpres230@ns1:~$ sudo debsums -c sl
/usr/games/sl
cpres230@ns1:~$ debsums sl
/usr/games/sl FAILED
/usr/share/doc/sl/README OK
/usr/share/doc/sl/README.Debian OK
/usr/share/doc/sl/README.jp OK
/usr/share/doc/sl/changelog.Debian.gz OK
/usr/share/doc/sl/copyright OK
/usr/share/man/de.UTF-8/man6/LS.6.gz OK
/usr/share/man/de.UTF-8/man6/sl.6.gz OK
/usr/share/man/de/man6/LS.6.gz OK
/usr/share/man/de/man6/sl.6.gz OK
/usr/share/man/ja.UTF-8/man6/LS.6.gz OK
/usr/share/man/ja.UTF-8/man6/sl.6.gz OK
/usr/share/man/ja/man6/LS.6.gz OK
/usr/share/man/ja/man6/sl.6.gz OK
/usr/share/man/man6/LS.6.gz OK
/usr/share/man/man6/sl.6.gz OK
cpres230@ns1:~$ _
```

- 3) Screenshots of four successful external forward lookups on your infrastructure (5 points)

```
cpre230@ns1:~$ dig ns1.student11.230.com +noall +answer
ns1.student11.230.com. 1543 IN A 42.49.30.200
cpre230@ns1:~$ dig www.student11.230.com +noall +answer
www.student11.230.com. 1800 IN A 42.49.30.202
cpre230@ns1:~$ dig mail.student11.230.com +noall +answer
mail.student11.230.com. 1800 IN A 42.49.30.204
cpre230@ns1:~$ dig www2.student11.230.com +noall +answer
www2.student11.230.com. 1800 IN A 42.49.30.206
cpre230@ns1:~$ _
```

- 4) Screenshots of four successful external reverse lookups on your infrastructure (5 points)

```
cpre230@ns1:~$ dig -x 42.49.30.200 +noall +answer
200.30.49.42.in-addr.arpa. 1442 IN PTR ns1.student11.230.com.
cpre230@ns1:~$ dig -x 42.49.30.202 +noall +answer
202.30.49.42.in-addr.arpa. 1800 IN PTR www.student11.230.com.
cpre230@ns1:~$ dig -x 42.49.30.204 +noall +answer
204.30.49.42.in-addr.arpa. 1526 IN PTR mail.student11.230.com.
cpre230@ns1:~$ dig -x 42.49.30.206 +noall +answer
206.30.49.42.in-addr.arpa. 1800 IN PTR www2.student11.230.com.
cpre230@ns1:~$ _
```

- 5) Screenshots of four successful external forward lookups on another student's Infrastructure

```
cpre230@ns1:~$ dig ns1.student108.230.com +noall +answer
ns1.student108.230.com. 900 IN A 73.19.46.200
cpre230@ns1:~$ dig www.student108.230.com +noall +answer
www.student108.230.com. 900 IN A 73.19.46.202
cpre230@ns1:~$ dig mail.student108.230.com +noall +answer
mail.student108.230.com. 863 IN A 73.19.46.204
cpre230@ns1:~$ dig www2.student108.230.com +noall +answer
www2.student108.230.com. 900 IN A 73.19.46.206
cpre230@ns1:~$
```

- 6) Screenshots of four successful external reverse lookups on another students Infrastructure. (5 points)

```
cpre230@ns1:~$ dig -x 73.19.46.200 +noall +answer
200.46.19.73.in-addr.arpa. 900 IN PTR ns1.student108.230.com.
cpre230@ns1:~$ dig -x 73.19.46.202 +noall +answer
202.46.19.73.in-addr.arpa. 900 IN PTR www.student108.230.com.
cpre230@ns1:~$ dig -x 73.19.46.204 +noall +answer
204.46.19.73.in-addr.arpa. 900 IN PTR mail.student108.230.com.
cpre230@ns1:~$ dig -x 73.19.46.206 +noall +answer
206.46.19.73.in-addr.arpa. 900 IN PTR www2.student108.230.com.
cpre230@ns1:~$ _
```

- 7) Screenshots of eight successful internal forward lookups on your personal infrastructure from your ns2 (10 points)

```
cpre230@ns2:/etc/bind$ dig ns2.student11.230.com +noall +answer
ns2.student11.230.com. 492 IN A 192.168.1.200
cpre230@ns2:/etc/bind$ dig desktop1.student11.230.com +noall +answer
desktop1.student11.230.com. 1800 IN A 192.168.1.201
cpre230@ns2:/etc/bind$ dig www.student11.230.com +noall +answer
www.student11.230.com. 1800 IN A 192.168.1.202
cpre230@ns2:/etc/bind$ dig mail.student11.230.com +noall +answer
mail.student11.230.com. 623 IN A 192.168.1.204
cpre230@ns2:/etc/bind$ dig ldap.student11.230.com +noall +answer
ldap.student11.230.com. 1800 IN A 192.168.1.205
cpre230@ns2:/etc/bind$ dig www2.student11.230.com +noall +answer
www2.student11.230.com. 1800 IN A 192.168.1.206
cpre230@ns2:/etc/bind$ dig ws.student11.230.com +noall +answer
ws.student11.230.com. 1800 IN A 192.168.1.207
cpre230@ns2:/etc/bind$ dig splunk.student11.230.com +noall +answer
splunk.student11.230.com. 1800 IN A 192.168.1.208
cpre230@ns2:/etc/bind$ _
```

- 8) Screenshots of eight successful internal reverse lookups on your personal infrastructure from your ns2
(10 points)

```
cpre230@ns2:~$ dig -x 192.168.1.200 +noall +answer
200.1.168.192.in-addr.arpa. 1800 IN PTR ns2.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.201 +noall +answer
201.1.168.192.in-addr.arpa. 1800 IN PTR desktop1.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.202 +noall +answer
202.1.168.192.in-addr.arpa. 1800 IN PTR www.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.204 +noall +answer
204.1.168.192.in-addr.arpa. 1800 IN PTR mail.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.205 +noall +answer
205.1.168.192.in-addr.arpa. 1800 IN PTR ldap.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.206 +noall +answer
206.1.168.192.in-addr.arpa. 1800 IN PTR www2.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.207 +noall +answer
207.1.168.192.in-addr.arpa. 1800 IN PTR ws.student11.230.com.
cpre230@ns2:~$ dig -x 192.168.1.208 +noall +answer
208.1.168.192.in-addr.arpa. 1800 IN PTR splunk.student11.230.com.
cpre230@ns2:~$
```

- 9) Take a screenshot of a forward lookup on another student's ldap machine (ldap.studentXX.230.com) from your ns1.
(10 points)

```
cpre230@ns1:/etc/bind$ dig ldap.student108.230.com
; <<>> DiG 9.18.30-Ubuntu0.22.04.2-Ubuntu <<>> ldap.student108.230.com
; global options: +cmd
; Got answer:
; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 47485
; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
; QUESTION SECTION:
; ldap.student108.230.com. IN A

; AUTHORITY SECTION:
student108.230.com. 1390 IN SOA ns1.student108.230.com. root.ns1.student108.230.com.
2025022402 604800 86400 2419200 604800

; Query time: 0 msec
; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
; WHEN: Mon Mar 03 06:28:21 UTC 2025
; MSG SIZE rcvd: 97
```

- 10) Did the query in question 9 resolve? Why or why not? Be specific.
(10 points)

Ans) The query did not resolve as we get the status as “NXDOMAIN”, which means that the domain does not exist. The other thing is that in the authority section we can see a “SOA”, which means start of authority. We only get that when the output is missing some specific information, for example, in this case the ldap query is missing a serial number and “ns1.student108.230.com root.ns1.student108.230.com.”

- 11) Screenshot of Nagios showing ns1 and n2 information all green
(20 points total, 10 points for ns1, 10 points for ns2)

Student 11 (student11)		
Host	Services	Actions
desktop1.student11.230.com	desktop1 ssh	  
ldap.student11.230.com	ldap Bezos ldap Gates ldap Jobs ldap Musk ldap Zuckerberg ldap service	  
mail.student11.230.com	dns MX mail pop secure imap secure imap login secure smtp	  
ns1.student11.230.com	dns mail dns ns1 dns www dns www2 reverse dns mail reverse dns ns1 reverse dns www reverse dns www2	  
ns2.student11.230.com	dns desktop1 dns ldap dns ws reverse dns desktop1 reverse dns ldap reverse dns ws	  
www.student11.230.com	www http www https www login	  
www2.student11.230.com	www2 django www2 ssh	  